MACHINE LEARNING IN PHYSICS
PHYSICS INFORMED NEURAL NETWORKS (PINN)

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Introduction

- The method of physics-informed neural networks (PINN) is a way[1,2] to solve ordinary and partial differential equations (ODE, PDE) using neural networks.

- The method leverages the modeling flexibility of neural networks and the computational tools that have been developed to fit enormously complicated functions.

1. Raissi, M., Perdikaris, P., Karniadakis, G.E., Physics-informed neural networks: A deep learning frame-work for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707 (2019).
2. Cuomo, S *et al*. Journal of Scientific Computing (2022) 92:88 https://doi.org/10.1007/s10915-022-01939-z

Introduction

Advantages of the PINN method

- Provides mesh-free solutions.
- Can incorporate data to guide solutions.
- Scales well with the dimensionality of the space.
- Can find solutions conditioned on initial/boundary conditions. In effect, solutions can be found for *infinitely* many initial/boundary conditions simultaneously.

Disadvantages of PINNs

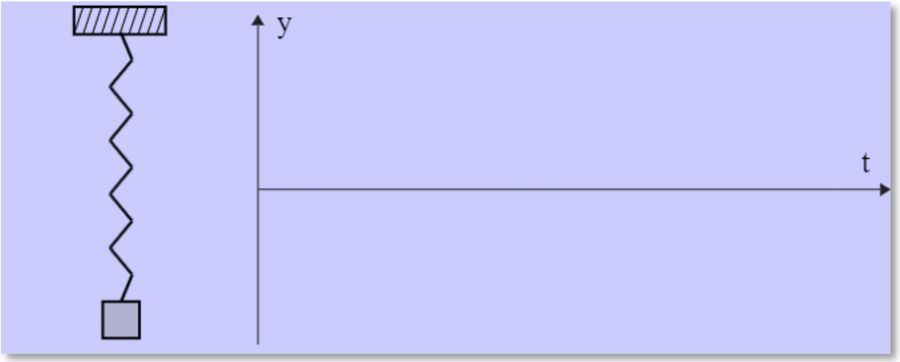
- For a given set of initial/boundary conditions, much slower than traditional numerical methods.
- Convergence is sensitive to the details of the average loss.

Introduction

The essential ideas of PINNs will be explained using the well-known problem of the damped harmonic oscillator,

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega_0^2 x = 0, \qquad \lambda = \frac{c}{m}, \omega_0 = \sqrt{\frac{k}{m}}$$

where x is the displacement of an object of mass m, k is the spring constant, c the viscous damping coefficient, and ω_0 the natural frequency of the spring.



https://www.mathwarehouse.com/harmonic-motion/interactive-damped-oscillator.php

Introduction

The dynamics of the damped harmonic oscillator can be described with a single free parameter $\zeta = \lambda/\omega_0$, the damping ratio, which allows us to write

$$\frac{d^2x}{dz^2} + 2\zeta\frac{dx}{dz} + x = 0$$

where the dimensional time is $z = \omega_0 t$.

This is a 2nd order ODE, therefore, to get a particular solution we need to specify initial conditions:

$$x(0) = x_0, \qquad \left.\frac{dx}{dz}\right|_{z=0} = v_0$$

Introduction

The damped harmonic oscillator can be solved exactly.

We'll consider the underdamped case $\zeta < 1$, for which the solution is

$$x(z) = \frac{x_0}{\cos\phi}\exp(-\zeta z)\cos\left(z\sqrt{1-\zeta^2}+\phi\right)$$

$$\phi = \arctan\left(-\frac{(\zeta + v_0/x_0)}{\sqrt{1-\zeta^2}}\right)$$

- The exact solution can be used to check the accuracy of the neural network solutions.
- We'll compute solutions *conditioned* on the triplet $(x_0, v_0, \zeta) \in S \subset \mathbb{R}^3$.

PHYSICS-INFORMED NEURAL NETWORKS (PINN)

Basic Idea (1)

1. Model the solution $x(z)$ with a neural network $f_\omega(z, p)$, where $p = (x_0, v_0, \zeta)$ and ω are the network parameters. How?

2. By minimizing an empirical risk (aka objective, or cost, function) comprising a *weighted sum* of three components:
    1. $R_{ode}(\omega)$ imposes physics constraint, via ODE;
    2. $R_{init}(\omega)$ imposes initial/boundary conditions,
    3. $R_{data}(\omega)$ imposes constraints provided by data.

Basic Idea (2)

The overall average loss function is a weighted sum of

$$R_{ode}(\omega) \;= \frac{1}{N_{ode}} \sum_{i=1}^{N_{ode}} \mathcal{F}^2[f_\omega(z_i, p_i)]$$

$$R_{init}(\omega)$$

$$= \frac{1}{N_{init}} \sum_{i=1}^{N_{init}} \left(\left(f_\omega(0, p_i) - x_{0,i}\right)^2 + \left[\frac{d f_\omega(0,\; p_i)}{dz} - v_{0,i}\right]^2 \right)$$

$$R_{data}(\omega) \;= \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} [f_\omega(z_i, p_i) - x(z_i)]^2$$

$\mathcal{F}\left(f_\omega(z_i, p_i)\right) = 0$ is the ODE and $p = (x_0, v_0, \zeta)$.

Problem of Weights

One problem with the average loss (aka objective function)

$$R(\omega) = w_{ode}\, R_{ode}(\omega) + w_{init}\, R_{init}\,(\omega) + w_{data}\, R_{data}(\omega)$$

is that minimization can be sensitive to the relative weights of the three components.

Moreover, since the different terms are in competition the accuracy of the solution also depends on the choice of weights.

Solution Ansatz

One way to reduce the sensitivity to the weights w_{ode}, w_{init}, and w_{data} is to eliminate w_{init} by incorporating the initial conditions into the solution explicitly.

For example, we could model the solution $x(z) \equiv f_\omega(z, p)$ using the *ansatz*[1]

$$x(z) = x_0 + g_\omega(z, p) - g_\omega(0, p) + z\,[v_0 - \dot{g}_\omega(0, p)]$$
$$\rightarrow \frac{dx}{dz} = v_0 + \dot{g}_\omega(z, p) - \dot{g}_\omega(0, p)$$

with $p = (x_0, v_0, \zeta)$ and $\dot{g} = dg/dz$.

1. D. Mortari, The Theory of Connections: Connecting Points, Mathematics, vol. 5, no. 57, 2017.

Automatic Differentiation

The key technology that makes the PINN method feasible is automatic differentiation (autodiff).

This is the technology that is used to compute the gradients, *exactly*, of the average loss with respect to the parameters ω of a neural network.

We can use autodiff to compute the derivatives with respect to z that appear in the average loss functions.

Autodiff: Example (1)

We can compute

$$\frac{\partial x}{\partial z}$$

using

```
dxdz = torch.autograd.grad
                 (x, z,
                  torch.ones_like(x),
                  create_graph=True)[0]
```

Autodiff: Example (2)

To compute

$$\frac{\partial^2 x}{\partial z^2}$$

we can do

```
d2xdz2 = torch.autograd.grad
              (dxdz, z,
               torch.ones_like(dxdz),
               create_graph=True)[0]
```

PINNs: Average Loss

Compute average loss $R(\omega)$.

First pick off ζ from $p = (x_0, v_0, \zeta)$, change its shape,

```
zeta = p[:,-1].view(-1, 1)
```

then compute $F = \frac{d^2x}{dz^2} + 2\zeta\frac{dx}{dz} + x$

```
F = d2xdz2 + 2*zeta*dxdz + x
R = torch.mean(F**2)
```

Summary

- Physics-informed neural networks offer a new way to solve differential equations in many scientific and engineering domains.

- Finding solutions with PINNs is typically more computationally demanding than with traditional numerical methods and convergence to solutions can be difficult.

- However, when the method is successful an infinite number of solutions can be found at the same time, each corresponding to a different set of initial/boundary conditions and problem parameters.